Business and System Requirements Specification (BRS and SRS) for "E-commerce System"

1. Introduction

The e-commerce system is designed to facilitate the buying and selling of products online. The system will handle the management of products, orders, customers, inventory, and categories, providing an efficient and user-friendly platform for both customers and administrators.

2. Stakeholders and Their Requirements

2.1. Stakeholders

1. **Customers**
    - **Role**: End-users who will browse, purchase products, and manage their profiles.
    - **Requirements**:
        - **Product Browsing**: Customers should be able to search, filter, and view products by categories, price, and stock availability.
        - **Order Management**: Customers should be able to place orders, view order status, and track shipments.
        - **Account Management**: Customers should be able to register, log in, and manage their personal information and order history.
        - **Payment**: Ability to make secure payments using various methods (credit/debit card, digital wallets, etc.).
        - **Reorder Notifications**: Notify customers when products in their order history are at risk of going out of stock.
2. **Administrators**
    - **Role**: System administrators who manage the backend, including product, order, and customer management.
    - **Requirements**:
        - **Product Management**: Administrators need the ability to add, update, and delete products, assign categories, and manage stock levels.
        - **Order Management**: Admins should be able to view, process, and update orders.

        - **Customer Management**: Admins should be able to manage customer accounts and address inquiries or issues.
        - **Inventory Management**: Ability to track stock levels, reorder products when stock reaches reorder levels.
        - **Sales and Revenue Reporting**: Ability to generate reports on sales, revenue, and popular products.
        - **Category and Warehouse Management**: Manage categories and assign products to specific warehouses.

3. **Warehouse Staff**
    - **Role**: Staff who manage product storage, stock levels, and shipments.
    - **Requirements**:
        - **Stock Monitoring**: Ability to view and manage stock quantities of products.
        - **Reorder Alerts**: Receive alerts when stock levels fall below reorder levels.
        - **Shipment Processing**: Track and update the status of product shipments to customers.
4. **Payment Gateway Provider**
    - **Role**: External service responsible for handling secure financial transactions.
    - **Requirements**:
        - **Secure Payment Processing**: Process payments and return success or failure status for transactions.
        - **Refund Management**: Handle refund requests for canceled orders or returns.
5. **Shipping Providers**
    - **Role**: External services that deliver products to customers.
    - **Requirements**:
        - **Shipping Management**: Ability to integrate with the system for shipment tracking and updating delivery statuses.
        - **Delivery Times**: Provide expected delivery times and handle delays or changes to delivery schedules.

3. Business Requirements

1. **Product Management**:
    - The system should allow administrators to create, update, and delete products, including attributes like name, price, stock quantity, and reorder level.
    - Products should be assigned to specific categories and warehouses.
    - Each product should have a unique identifier (ProductID), and stock should be tracked to ensure inventory levels are accurate.
2. **Order Management**:
    - Customers should be able to view available products, add them to the cart, and complete the purchase process.
    - The system should handle order status updates, including processing, shipped, delivered, and canceled statuses.
    - Administrators should be able to view and process orders, managing payment and shipment details.
3. **Customer Management**:
    - Customers should have accounts to manage their information and track order history.
    - The system should allow customers to change personal information, view past orders, and manage returns or refunds.
4. **Shipping and Delivery**:
    - The system should integrate with shipping providers to calculate delivery costs, track orders in transit, and update customers on expected delivery dates.
    - Customers should be able to track their shipments through the platform.
5. **Inventory Management**:
    - The system should track product stock levels and notify warehouse staff when products reach reorder levels.
    - Administrators should be able to adjust stock levels manually or automatically based on incoming shipments or order cancellations.
6. **Payment Processing**:

- The system must securely process payments using various methods (credit cards, digital wallets, etc.).
- Payment transactions must be recorded and linked to the corresponding customer order.
- The system should handle refunds for canceled orders or returns.

7. **Reporting and Analytics**:
   - The system should provide reports for administrators on sales, revenue, popular products, and customer activity.
   - Reports should help track business performance and identify trends.

4. System Requirements

4.1. Functional Requirements

1. **User Authentication**:
   - The system should provide login and registration functionality for customers and administrators.
2. **Product Search and Filtering**:
   - Customers should be able to search for products by name, category, price, and stock availability.
3. **Shopping Cart**:
   - Customers should be able to add items to their cart, modify quantities, and remove items before proceeding to checkout.
4. **Order Processing**:
   - The system should capture customer information, payment details, and shipping addresses to process orders.
5. **Admin Panel**:
   - Administrators should have access to a backend panel to manage products, orders, and customers.
6. **Payment Integration**:
   - The system should integrate with a secure payment gateway for processing financial transactions.
7. **Shipping Integration**:

- The system should integrate with external shipping providers to track and update delivery statuses.
- **CREATE TABLES**

*CREATE TABLE Customers (

CustomerID INT PRIMARY KEY,

FirstName VARCHAR(50),

LastName VARCHAR(50),

Email VARCHAR(100) UNIQUE,

PhoneNumber VARCHAR(20),

ShippingAddress TEXT

);

*CREATE TABLE Orders (

OrderID INT PRIMARY KEY,

CustomerID INT,

OrderDate DATETIME DEFAULT CURRENT_TIMESTAMP,

Status VARCHAR(50) DEFAULT 'Pending',

TotalAmount DECIMAL(10, 2),

FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)

);

*CREATE TABLE Category (

CategoryID INT PRIMARY KEY,

Name VARCHAR(255) NOT NULL,

DiscountPlanName VARCHAR(255),

DiscountPercentage DECIMAL(5, 2),

StartDate DATE,

EndDate DATE

```
);

*CREATE TABLE OrderDetails (
   OrderDetailID INT  PRIMARY KEY,
   OrderID INT,
   ProductID INT,
   Quantity INT NOT NULL,
   Subtotal DECIMAL(10, 2),
   FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
   FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);
 *CREATE TABLE Cart (
   CartID INT  PRIMARY KEY,
   CustomerID INT,
   Date DATE,
   Day INT,
   Month INT,
   Year INT,
   FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);
*CREATE TABLE Payments (
   PaymentID INT  PRIMARY KEY,
   PaymentType VARCHAR(50),
   Date DATE,
   Day INT,
   Month INT,
   Year INT,
   Status VARCHAR(50),
```

```
    OrderID INT,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
);
*CREATE TABLE Warehouses (
    WarehouseID INT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    Location VARCHAR(255),
    ContactInfo VARCHAR(255),
    Manager VARCHAR(255)
);
*CREATE TABLE Delivery (
    DeliveryID INT  PRIMARY KEY,
    DeliveryMan VARCHAR(255),
    Status VARCHAR(50),
    OrderID INT,
    TrackingURL VARCHAR(255),
    ShippingProviderID INT,
    ActualDeliveryDate DATE,
    EstimatedDeliveryDate DATE,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    FOREIGN KEY (ShippingProviderID) REFERENCES
ShippingProviders(ShippingProviderID)
);
*CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    Name VARCHAR(255) NOT NULL,
    Price DECIMAL(10, 2) NOT NULL,
    StockQuantity INT NOT NULL,
```

```
    ReorderLevel INT NOT NULL,
    CategoryID INT,
    WarehouseID INT,
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID),
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID)
);
```

4.2. Non-Functional Requirements

1. **Performance**:
    - The system should handle a large number of concurrent users and transactions without significant delays.
2. **Scalability**:
    - The system should be scalable to handle increased traffic during peak times (e.g., holidays or sales events).
3. **Security**:
    - The system must use secure communication protocols (SSL/TLS) for data transmission, particularly for payment transactions.
    - Passwords must be stored securely using hashing algorithms.
4. **Usability**:
    - The user interface (UI) should be intuitive and easy to navigate for both customers and administrators.
5. **Reliability**:
    - The system should have a high level of uptime, with minimal maintenance disruptions.
6. **Backup and Recovery**:
    - The system should have automated backup procedures in place to recover from data loss.
7. **Compliance**:
    - The system should comply with relevant data protection regulations (e.g., GDPR for customer data).
